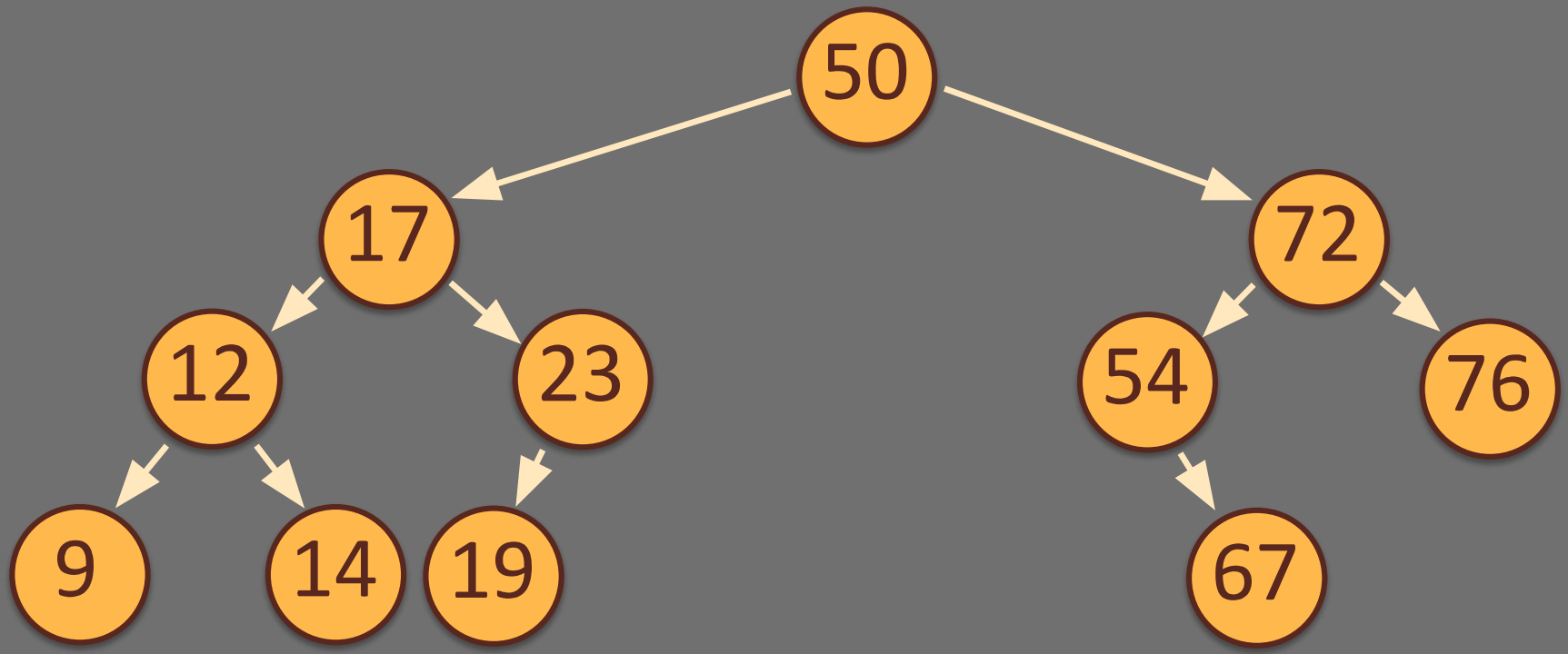
B+-Tree Indices

Review: Balanced search tree (BST)



B+-tree

BST optimized for disk-based access:

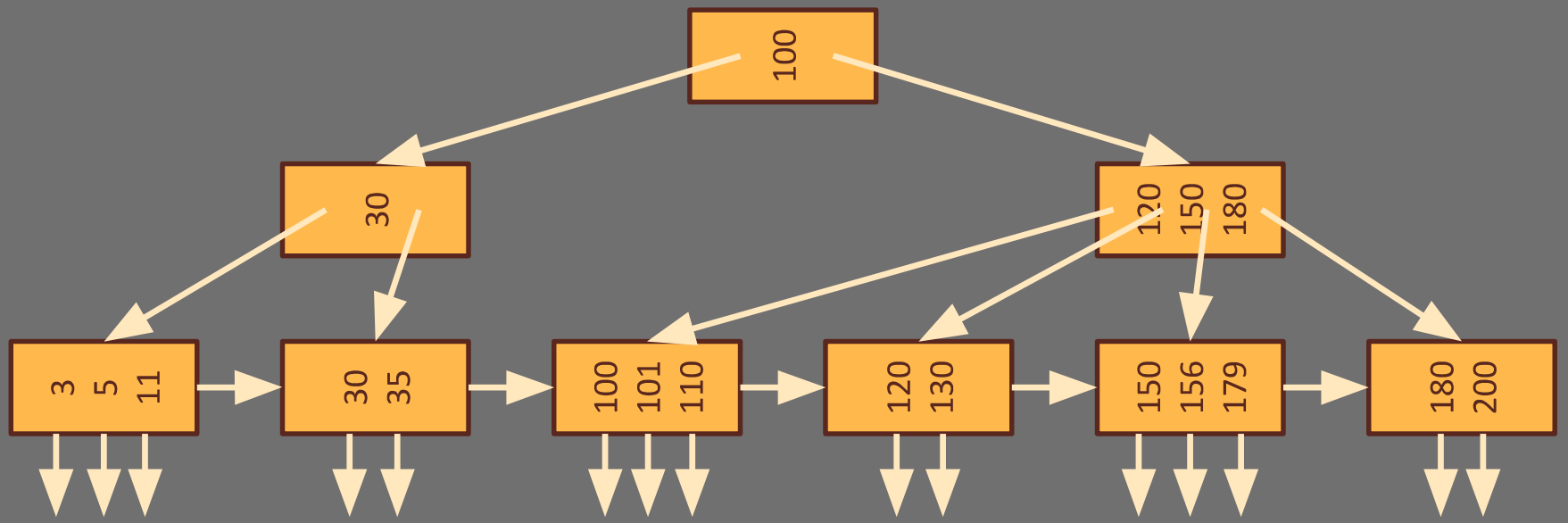
- Shallow
- Wide: typical node size = page.

Properties for indexing:

- All data in leaf nodes
- Leaf nodes sequentially linked

} Can traverse data in order.

B+-tree example



… Data records

…

Search key uniqueness

Supports uniqueness & non-uniqueness.

- Unique: $<$ order
- Non-unique: $\leq$ order

Indexing on text search key

Text length varies.

- #values in a node varies.
- Long text reduces #children (fan-out).

Techniques to maximize fan-out:

- Internal nodes store only prefixes of search key values.
  E.g., “Johnson” and “Jones” can be separated by “Jon”.

- Leaf nodes share common prefixes.
  E.g., “Johnson” and “Jones” can be stored as “Jo” + “hnson”/”nes”

Index operations

- Search, Insert, Delete: $O(\log n)$
  - Details irrelevant
- Can linearly traverse tuples after search
  - Return first or all matching tuples.
  - Find search key in a range of values.

- No fragmentation of index